A380X fuel jettison and FINAL fuel

Fuel jettison down to the JTSN GW with its ECAM memo and procedures, the jettison valve failures with their alerts, a fuel mist plume at the jettison outlets, the FINAL fuel and time from the A380 holding fuel flow, and TRIP and cruise fuel predictions that match the fuel the engines burn

Aircraft	A380X
Where to find it	Overhead FUEL panel: JETTISON ARM and ACTIVE; MFD FMS ACTIVE/FUEL&LOAD: JTSN GW; E/WD and SD FUEL; flyPad > Failures > ATA 28; outside view of the wings; MFD FMS FUEL&LOAD: FINAL and EXTRA time, TRIP fuel; F-PLN EFOB predictions
Branch	`core/fuel`, based on `core/a380x-mfd-fms` (stacked: needs `core/surveillance` and `core/a380x-mfd-fms` merged first). The plume needs the aircraft-large-files companion change fa6c5d1 (jettison outlet nodes, fork TCdr/aircraft-large-files).
Status	Jettison: tested in the sim (September 2026), rate checked live on 2026-10-02 (2.5 t/min, feed tanks untouched). Valve failures with FUEL JETTISON FAULT, VLV NOT CLOSED and INOP SYS JETTISON: OK 2026-10-02. Plume: look and logic OK in flight 2026-10-03, position at the outlets OK 2026-10-04. SD FUEL jettison lines white during a normal jettison: sim-tested on develop 2026-10-04. FINAL fuel and time from the holding fuel flow: OK at the gate on develop 2026-10-08 (user word); its EXTRA time: not sim-tested. TRIP and cruise fuel predictions: tested in the sim by the user on the integration build before the commit, climb and 20 min cruise at M.85 (tests T1/T2, 2026-10-09): OK. The core branch itself was not sim-tested.
Pictures	MFD and EWD in the local browser harness, running the builds installed in the sim; the EWD states are the ones the FWS sends in flight. No pictures for the failures and the plume.

1. Fuel jettison with the JTSN GW, ECAM memo and procedures
2. Jettison valve failures: FUEL JETTISON FAULT, VLV NOT CLOSED, INOP SYS JETTISON
3. Fuel jettison plume at the outlets
4. FINAL fuel and time from the A380 holding fuel flow
5. TRIP and cruise fuel predictions match the engines

1. Fuel jettison with the JTSN GW, ECAM memo and procedures

> **Aircraft:** A380X | **Where:** Overhead JETTISON ARM / ACTIVE; MFD FUEL&LOAD JTSN GW; ECP ABN PROC | **Status:** Tested in the sim by the user (September 2026); jettison rate measured live on 2026-10-02: 2.5 t per minute from the outer, mid and trim tanks, feed tanks untouched; SD FUEL jettison lines white during a normal jettison: sim-tested on develop 2026-10-04. Unit tests: FuelJettison.spec.ts (FCOM DSC-28-40), JettisonValveIndication.spec.ts (SD colours, DSC-28-20).

What it is for

FBW had the overhead JETTISON ARM and ACTIVE pushbuttons, but they did nothing. The fuel management (FQMS) now jettisons the fuel of the transfer tanks down to the gross weight entered on the MFD, and the ECAM shows the memo and the procedures of the FCOM.

1. Set the jettison gross weight

1. MFD FMS **ACTIVE/FUEL&LOAD** page. **JTSN GW** is shown when the active flight plan has its mandatory data.
2. Enter the gross weight at which the jettison must stop, in tonnes (or 1000 lb with the LBS unit).
3. Allowed range: from **ZFW + 11 000 lb** (5.0 t) to the maximum gross weight (FCOM DSC-22-FMS-20-100). Outside it: ENTRY OUT OF RANGE with the range.

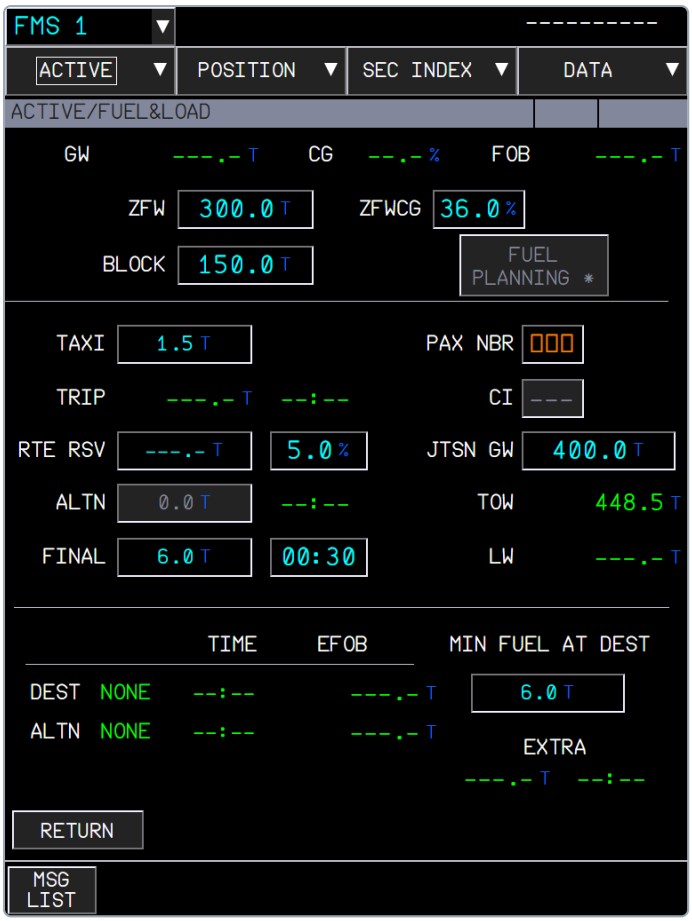


Figure 1. JTSN GW 400.0 t entered (ZFW 300.0 t).

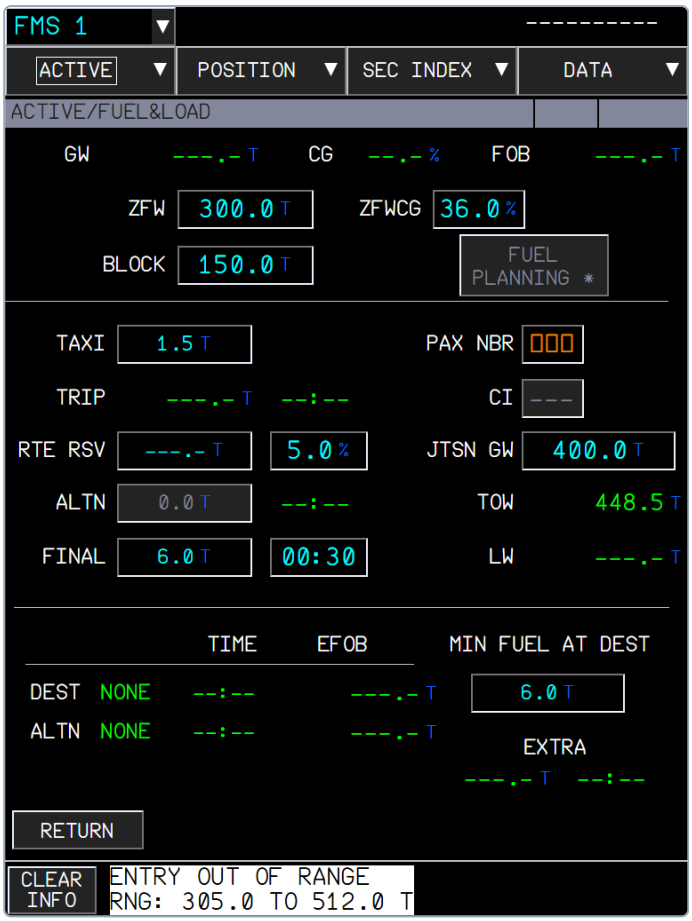


Figure 2. 290 t is below ZFW + 5.0 t: the range is shown in the message area.

> **CAUTION**
> Without a JTSN GW the FQMS has no target weight: the jettison then runs until the transfer tanks are empty (or a pushbutton is set OFF).

2. Run the procedure

The ABN PROC procedure takes the crew through the steps. ECP **ABN PROC**, then FUEL, then JETTISON:

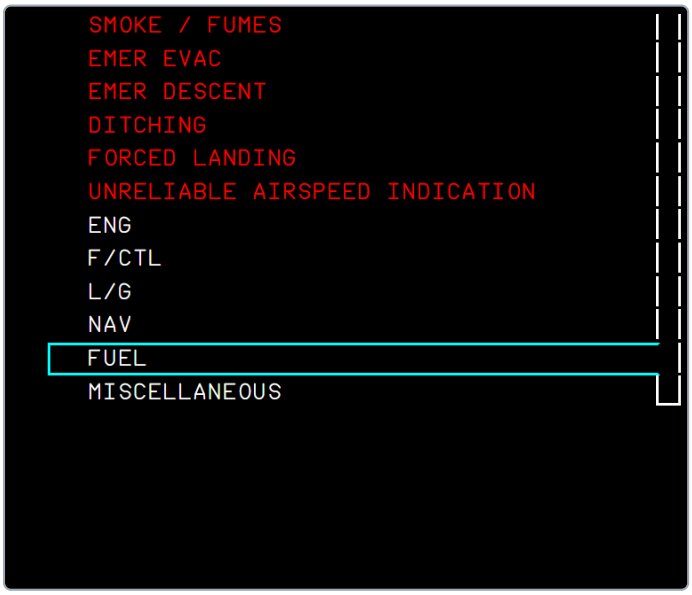


Figure 3. ABN PROC menu: FUEL selected.

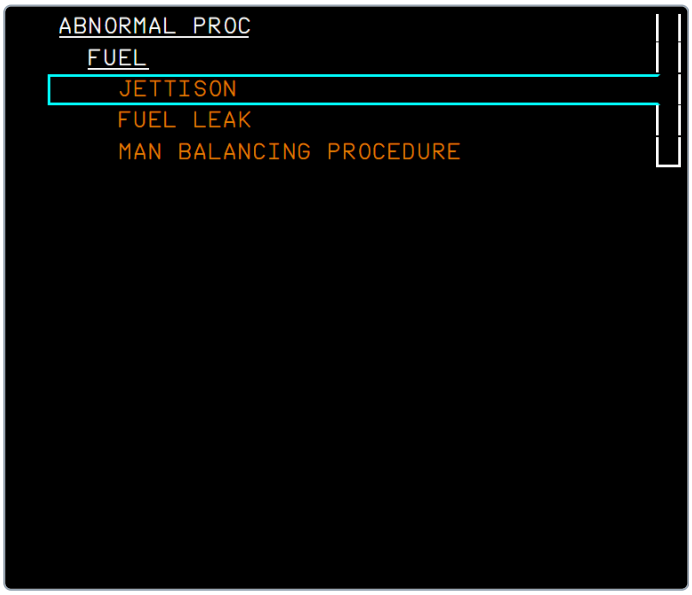


Figure 4. FUEL sub-menu: JETTISON.

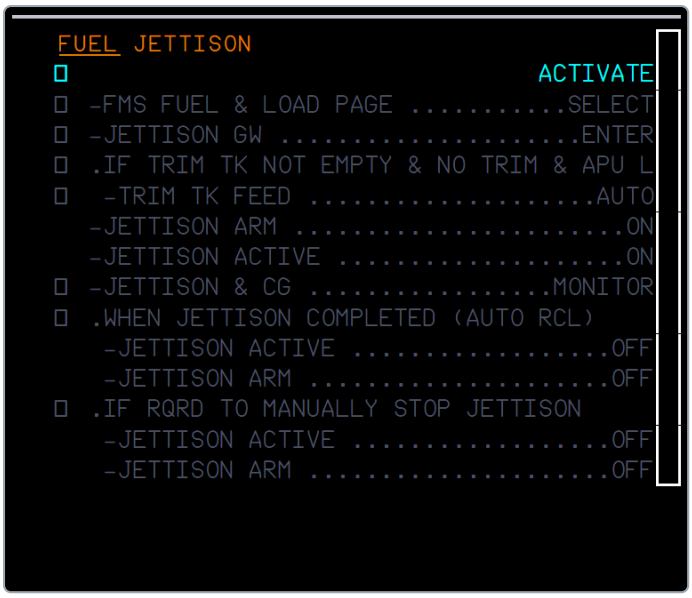


Figure 5. The procedure before ACTIVATE: all lines in grey.

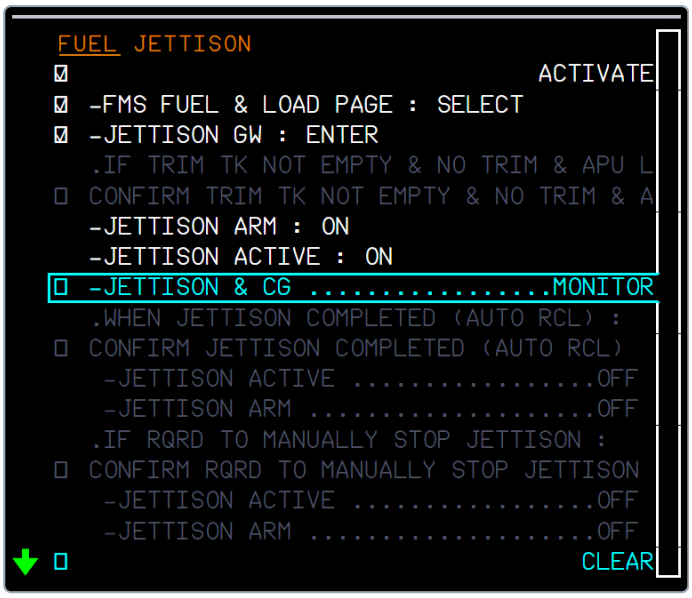


Figure 6. Activated: the FMS lines checked by the crew, JETTISON ARM and ACTIVE ticked by the FWS as the pushbuttons are set (sensed lines).

- FMS FUEL & LOAD PAGE ... SELECT, JETTISON GW ... ENTER: crew check lines.
- IF TRIM TK NOT EMPTY & NO TRIM & APU LINE FAULT: TRIM TK FEED AUTO (condition line, confirmed by the crew).
- **JETTISON ARM ON**, **JETTISON ACTIVE ON**: sensed, they tick themselves.
- JETTISON & CG ... MONITOR: watch the SD FUEL page and the CG.
- WHEN JETTISON COMPLETED (AUTO RCL) / IF RQRD TO MANUALLY STOP JETTISON: ACTIVE OFF then ARM OFF, sensed.

3. During the jettison

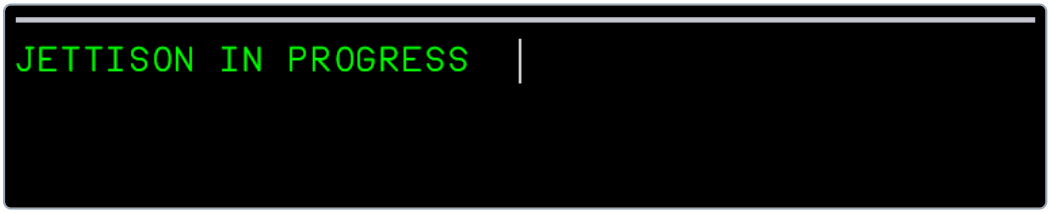


Figure 7. The EWD memo while the fuel is jettisoned.

- Memo **JETTISON IN PROGRESS** (green).
- Fuel leaves the **transfer tanks** (outer, mid, inner and trim) at the FCOM rate of **330 693 lb/h** (about 2.5 t per minute), in proportion to their content, so they empty together. The **feed tanks are never jettisoned**.
- The trim tank is held when the CG reaches the forward T.O/LDG limit, and resumes when the CG is 0.5 % aft of it.
- The other automatic transfers pause during the jettison.
- The SD FUEL page shows the two jettison valves open, with the jettison lines and JETTISON legends in white (normal, FCOM DSC-28-20), and the quantities decreasing. Before the fix on this branch they were amber during every jettison (the page did not know a jettison was running).

4. When it stops

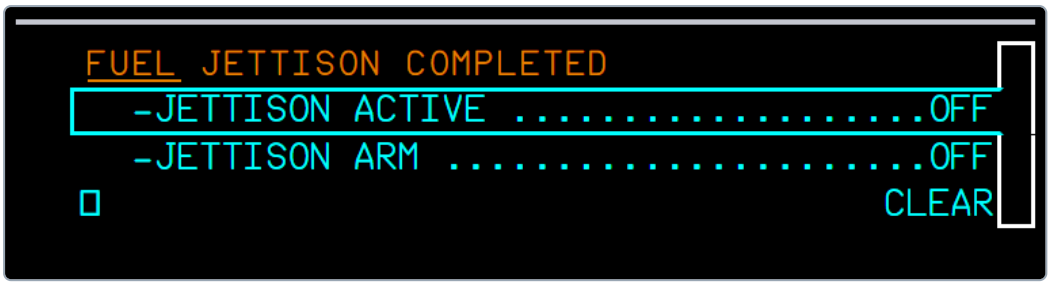


Figure 8. FUEL JETTISON COMPLETED: set ACTIVE then ARM OFF; the lines tick as the pushbuttons are released.

Situation	What you should see
Gross weight reaches the JTSN GW	The jettison stops, FUEL JETTISON COMPLETED shows, the ACTIVE ON light goes off.
All transfer tanks empty (or no JTSN GW entered)	Same: stops and COMPLETED.
ACTIVE or ARM set OFF by the crew	Stops at once (manual stop). No COMPLETED alert.
Both pushbuttons OFF after COMPLETED	The alert and the COMPLETED state clear; a new jettison can be started.
ACTIVE ON without ARM	Nothing: ARM is needed.
AC 1 bus not powered	No jettison (FQMS channel unpowered).

Limitations and why

Limitation	Why	What to do
JETTISON FAULT, VLV NOT CLOSED and INOP SYS JETTISON come only from the flyPad valve failures.	No other cause of a jettison valve fault is simulated; the four valve failures are described in the next feature.	-
The fuel does not really flow out through the outlets.	The MSFS fuel system has no overboard dump: the FQMS sets the tank quantities directly and opens valves 57 / 58 for the SD picture. A fuel mist plume is drawn at the outlets (last feature).	-
After an update, the first flight needs a full MSFS restart.	The change touches the behaviour XML of the overhead, which MSFS loads at start.	Restart MSFS once after installing.

How to use

1. MFD FMS **ACTIVE/FUEL&LOAD**: enter **JTSN GW** (the gross weight at which the jettison stops).
2. ECP **ABN PROC** > FUEL > JETTISON, ACTIVATE, and follow the lines.
3. Overhead FUEL panel: **JETTISON ARM** ON, then **ACTIVE** ON. Monitor the SD FUEL page and the CG.
4. At FUEL JETTISON COMPLETED (or to stop earlier): ACTIVE OFF, then ARM OFF.

How to test

#	Do this	Expected result
1	A380X with fuel in the transfer tanks (e.g. 150 t block), AC power, active flight plan with its mandatory data. MFD FMS ACTIVE/FUEL&LOAD: type a JTSN GW below ZFW + 5.0 t.	ENTRY OUT OF RANGE with the allowed range in the message area.

2	Type a JTSN GW about 20 t below the current gross weight.	JTSN GW shows the value.
3	ECP ABN PROC > FUEL > JETTISON, ACTIVATE.	The ABN PROC FUEL JETTISON procedure; JETTISON ARM and ACTIVE lines not ticked.
4	Set **JETTISON ACTIVE** ON without ARM.	Nothing happens (ARM is needed).
5	Set **JETTISON ARM** ON (ACTIVE ON).	Memo JETTISON IN PROGRESS (green); the ARM and ACTIVE lines tick; SD FUEL: both jettison valves open, jettison lines and JETTISON legends white (not amber), the outer, mid, inner and trim quantities fall together (about 2.5 t per minute); feed tanks unchanged.
6	Wait until the gross weight reaches the JTSN GW.	The jettison stops; FUEL JETTISON COMPLETED with ACTIVE OFF / ARM OFF; the ACTIVE ON light goes off.
7	Set ACTIVE OFF, then ARM OFF.	The lines tick and the COMPLETED alert clears.
8	Start a new jettison and set ACTIVE OFF halfway.	It stops at once, without COMPLETED.

2. Jettison valve failures: FUEL JETTISON FAULT, VLV NOT CLOSED, INOP SYS JETTISON

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28; E/WD; SD STATUS; overhead JETTISON ACTIVE OPEN light | **Status:** Tested in the sim by the user: one valve stuck closed, both stuck closed, stuck open, all OK (2026-10-02, after the FAULT / INOP SYS fix); the cases again with the plume on 2026-10-03. Unit tests: FuelJettison.spec.ts, FuelJettisonFlags.spec.ts.

What it is for

The A380 FCOM has two jettison alerts, FUEL JETTISON FAULT and FUEL JETTISON VLV NOT CLOSED (PRO-ABN-ECAM-10-28 P 61-63), and INOP SYS JETTISON on the STATUS page. FBW had no jettison valve failure to trigger them. There are now four failures, one per valve and position. A first version never showed the FAULT caution and the INOP SYS: the FWS read the FQMS flags as numbers (1) and combined them with a function that only counts `true`; they are now read as real booleans.

The failures

flyPad > Failures > ATA 28 (Fuel)	Effect
Jettison valve L stuck closed / **Jettison valve R stuck closed**	That valve does not open: the fuel goes through the other one at **half the rate**. The FQMS records the valve fault when the jettison is selected (ARM and ACTIVE ON), until the failure is removed.
Both stuck closed	The jettison is **not available**: no fuel leaves.
Jettison valve L stuck open / **Jettison valve R stuck open**	That valve stays open: the **OPEN** light of the JETTISON ACTIVE pushbutton stays on and the SD shows the valve open, also with no jettison running. No fuel is lost by it (the jettison itself still needs ARM and ACTIVE).

What you should see

Situation	What you should see
One valve stuck closed, jettison selected	Amber caution **FUEL JETTISON FAULT**; on the ground with the line JETTISON L (R) VLV FAULT. Jettison at half rate. STATUS: INOP SYS **JETTISON** (the FWS lists it whenever FUEL JETTISON FAULT is active).

Both valves stuck closed, jettison selected	**FUEL JETTISON FAULT** with JETTISON NOT AVAIL, then JETTISON ACTIVE ... OFF and JETTISON ARM ... OFF (they tick as the pushbuttons are set OFF). STATUS: INOP SYS **JETTISON**. No fuel jettisoned.
A valve stuck open, no jettison running	Amber caution **FUEL JETTISON VLV NOT CLOSED** with JETTISON ACTIVE ... OFF and JETTISON ARM ... OFF; OPEN light on.
Takeoff and landing	The alerts are inhibited in the takeoff and landing flight phases (FCOM inhibitions).

How to use

1. flyPad > **Failures** > **ATA 28**: activate one of the four jettison valve failures.
2. Run a jettison (or not, for a valve stuck open) and follow the ECAM.
3. Deactivate the failure in the flyPad to repair the valve.

How to test

#	Do this	Expected result
1	A380X on the ground (or in cruise), fuel in the transfer tanks, AC power, JTSN GW entered. flyPad > Failures > ATA 28: activate **Jettison valve L stuck closed**. Set JETTISON ARM and ACTIVE ON.	Amber **FUEL JETTISON FAULT** (on the ground with JETTISON L VLV FAULT); SD FUEL: only the right jettison valve open; the transfer tanks empty at half the normal rate (about 1.25 t per minute). STATUS: INOP SYS JETTISON.
2	Also activate **Jettison valve R stuck closed**.	FUEL JETTISON FAULT with JETTISON NOT AVAIL, JETTISON ACTIVE OFF, JETTISON ARM OFF; no fuel leaves; STATUS: INOP SYS JETTISON.
3	Set ACTIVE OFF, then ARM OFF.	The ACTIVE and ARM lines tick.
4	Deactivate both failures, activate **Jettison valve L stuck open** (pushbuttons OFF).	Amber **FUEL JETTISON VLV NOT CLOSED**; OPEN light of JETTISON ACTIVE on; SD FUEL: left jettison valve open; the fuel quantities do not change.
5	Deactivate the failure.	The valve closes, the OPEN light goes off and the alert clears.

3. Fuel jettison plume at the outlets

> **Aircraft:** A380X | **Where:** Outside view: wing trailing edge, at the second-outermost flap track fairing | **Status:** Tested in the sim by the user in flight: look, both / left only / right only and the valve failure cases OK (2026-10-03); position at the outlets OK (2026-10-04).

What it is for

The jettisoned fuel left no trace outside. A fuel mist plume now trails from each wing's jettison outlet while fuel flows out of that side. The FCOM says there is "one discharge point on the trailing edge of each wing" (DSC-28-40); the outlet is the tube at the aft end of the second-outermost flap track fairing (MFTF 5, forum source), where the companion model adds the effect points.

What you should see

Situation	What you should see
Jettison in progress, both valves open	A mist plume behind each wing outlet; it gets longer with the true airspeed (its particles stay where they were released).

One valve stuck closed	The plume on the other side only.
Valve stuck open, no jettison running	No plume (no fuel flows).
Below 40 kt true airspeed	The low-speed version of the mist (the stock MSFS water-ballast effect switches at 40 kt).

Limitations and why

Limitation	Why	What to do
Needs the aircraft-large-files companion.	The effect hangs on the FX_JETTISON_LH / RH nodes added at the outlets in the A380_WINGS models (fa6c5d1); without them no plume shows.	Install it; a full MSFS restart is needed after the update.
The look is the stock water-ballast mist.	Closest stock MSFS effect: design choice.	-
No fuel puddle on the ground.	Jettisoned fuel evaporates; nothing is drawn on the ground.	-

How to use

1. Run a fuel jettison (first feature) and look at the wings from the outside view.

How to test

#	Do this	Expected result
1	A380X in flight (companion models installed, MSFS restarted), JTSN GW entered. Set JETTISON ARM and ACTIVE ON. Outside view of a wing.	A mist plume trails from the trailing edge at the second-outermost flap track fairing, on both wings.
2	Speed up or slow down.	The plume gets longer at a higher true airspeed.
3	flyPad > Failures > ATA 28: **Jettison valve L stuck closed**.	The left plume stops; the right one goes on.
4	Set ACTIVE OFF.	Both plumes stop.
5	Remove the failure, activate **Jettison valve R stuck open** without a jettison.	No plume.

4. FINAL fuel and time from the A380 holding fuel flow

> **Aircraft:** A380X | **Where:** MFD FMS ACTIVE/FUEL&LOAD: FINAL fuel and time, EXTRA time | **Status:** FINAL fuel and time: tested in the sim by the user on develop at the gate: OK (2026-10-08, optional A380X card T1: ZFW 342.6 t, FINAL entry, ZFW 300.0 t; user word, no simvar to record). The EXTRA time: not sim-tested. Unit tests: FinalHoldingPredictor.spec.ts. The core branch itself was not sim-tested.

What it is for

The FINAL fuel default of the FUEL&LOAD page was a fixed 6.0 t per 30 min (0.2 t per minute, the same flow for the EXTRA time), whatever the weight. The A380 FCOM defines the holding fuel as 30 min at 1 500 ft above the alternate, at holding speed (green dot + 25 kt, CONF CLEAN), ISA, with the landing weight at the alternate (PER-IFT-FPL-MRF HOLDING FUEL). The FINAL fuel now comes from that holding.

- The fuel flow comes from the one A380 holding computation in the FCOM (PER-IFT-HLD-HMI, IN-FLT PERF example: 940.0 klb, 30 min, 13 100 lb), brought back from ISA+10 to ISA, and taken proportional to the weight.
- FINAL fuel: a holding of the FINAL time (30 min by default) that ends at the ZFW, rounded up to the 0.1 t of the entry field, like the ALTN fuel. Check: RJTT-RKSI at ZFW 342.6 t gives 4.70 t, shown 4.8 (the airline OFP gives 4.8 t).

- FINAL time from a FINAL fuel entry: the time that fuel lasts in the same holding.
- EXTRA time: the time the EXTRA fuel lasts in the same holding, ending with the ZFW plus the MIN FUEL AT DEST on board.
- Without a ZFW: the FCOM conservative quantity, 10 600 lb per 30 min.
- This replaces the fixed 0.2 t per minute flow (30 min = 6 t) of `core/a380x-mfd-fms`.

What you should see

Situation	What you should see
RJTT/RKSI, ALTN RKSS, ZFW 342.6 t, no FINAL entry	FINAL 4.8, time 00:30.
ZFW 300.0 t	FINAL 4.2, time 00:30.
FINAL fuel 6.0 entered (ZFW 342.6 t)	FINAL time 00:38.
FINAL fuel cleared (CLR)	Back to FINAL 4.8, 00:30.
5.0 t of EXTRA fuel at ZFW 342.6 t	EXTRA time about 00:30 or 00:31 (about 6.3 min per tonne; before: 5 min per tonne).

Limitations and why

Limitation	Why	What to do
The FINAL time stays 00:30 with a calculated fuel.	Design choice: the time shown is the one the fuel was computed from, not the slightly longer time of the rounded-up fuel.	-
The holding altitude and the temperature are not used.	The A380 FCOM has no holding table; the A320neo table varies by less than 5 % from FL15 to FL140, and the FCOM figure is brought back to ISA (design choice).	-
The fuel flow is taken proportional to the weight.	Maximum endurance: thrust follows the weight; the A320neo holding table confirms the near linear law (design choice).	-

How to use

1. MFD FMS **INIT**: FROM/TO and ALTN. **FUEL&LOAD**: enter the ZFW and ZFWCG.
2. Read the FINAL fuel and time. Overwrite the FINAL fuel or the FINAL time if the company policy asks for another value; CLR returns to the calculated value.
3. Read the EXTRA time next to the EXTRA fuel once the BLOCK is entered.

How to test

#	Do this	Expected result
1	A380X at the gate, aircraft powered, no flight plan. MFD (FMS) INIT: FROM/TO **RJTT/RKSI**, ALTN **RKSS**. FUEL&LOAD: ZFW **342.6**, ZFWCG **30.0**.	FINAL **4.8**, time **00:30** (before: 6.0).
2	Change the ZFW to **300.0**.	FINAL **4.2**, time 00:30. Set the ZFW back to 342.6.
3	Enter **6.0** in the FINAL fuel field.	FINAL time **00:38** (before: 00:30).
4	CLR the FINAL fuel.	FINAL back to **4.8**, time 00:30.

5	Run FUEL PLANNING, then add 5 t to the BLOCK (not sim-tested yet).	EXTRA time about **00:30** or **00:31** for 5.0 t (before: 00:25).

5. TRIP and cruise fuel predictions match the engines

> **Aircraft:** A380X (the shared VNAV engine model also changes the A32NX predictions) | **Where:** MFD FMS FUEL&LOAD TRIP; F-PLN EFOB and DEST EFOB | **Status:** Tested in the sim by the user on the integration build before the commit: climb FL100-FL300 and 20 min cruise at M.85 (test card test-card-a380-trip-fuel-sim-burn, T1/T2, 2026-10-09): OK. Unit tests: EngineModel.spec.ts, A380TripFuelPrediction.spec.ts. The A32NX predictions: not sim-tested.

What it is for

The A380X FMS predicted less fuel than the engines burn: 4 to 7 % less in cruise and about 3 % less on the TRIP (e.g. 19.8 t instead of 20.5 t on a 700 NM, FL340 RJTT-RKSI-like leg), so the DEST EFOB fell during the cruise. The cause was in the VNAV engine model: to find the N1 that gives a thrust, it inverted the engine thrust table (table 1506 of engines.cfg) column by column, which is not the inverse of the table the sim engine reads (bilinear in N1 and Mach). At M.85, exactly between the M.8 and M.9 columns, it took up to 2 % N1 too little. It now interpolates the table at the Mach first and inverts that single curve.

- The cruise fuel flow now matches the FADEC fuel flow at M.85 (A380TripFuelPrediction.spec.ts: the sim burn and the FCOM IN-FLT PERF examples).
- The same RJTT-RKSI-like leg (ZFW 342.6 t, 30 t at take-off, FL340, ISA, no wind) now gives a TRIP of 20.5 t.
- **A32NX:** `EngineModel.ts` is the shared FMGC code, so the A32NX VNAV predictions (climb, cruise, descent) use the corrected inverse too. They change only between the Mach columns of the A32NX thrust table.

Limitations and why

Limitation	Why	What to do
The TRIP is still below a SimBrief plan (23.5 t for the real RJTT-RKSI route).	The sim A380X itself burns less than the FCOM IN-FLT PERF examples (climb about 24 % less fuel, cruise about 5 % less); the FMS now matches the sim engines, not the FCOM tables.	-

How to use

1. Nothing to set: enter the flight plan, the CRZ FL and the FUEL&LOAD data as usual and read the TRIP and the EFOB predictions.

How to test

#	Do this	Expected result
1	A380X departed from RJTT with an RJTT-RKSI flight plan, MFD INIT CRZ FL 300, FUEL&LOAD done. In flight at FL100, clean, 250 kt, AP and A/THR ON, MSFS weather clear skies (ISA, no wind). FCU: altitude 30000, OP CLB, speed 300 kt.	The aircraft climbs and levels off at FL300.
2	FCU: push the SPD knob (managed M.85). After 3 min read the DEST EFOB on the MFD F-PLN page; read it again 20 min later.	The two DEST EFOB readings differ by at most 0.1 t (before the fix it fell by about 0.2 t).